

Recovering DATA

Proyecto de Exposición e Investigación

Recovering Deleted Files & Deleted Partitions

- How file systems handle deletion (logical vs. physical deletion, metadata vs. content areas).
- Common locations and artifacts for “deleted” data (recycle bin, unallocated space, slack space, volume shadow copies).
- File-recovery techniques: undelete via metadata, file carving/signature analysis, timeline-based recovery.
- Partition-level concepts: MBR/GPT, partition tables, volume structures, and how partitions are “deleted.”
- Recovering deleted or lost partitions with specialized tools and manual reconstruction.
- Challenges with SSDs and TRIM, wear-leveling, and secure deletion utilities.
- Anti-forensics techniques (wiping, encryption, obfuscation) and their impact on recovery.
- Validation and documentation of recovered data for evidentiary use (hashing, chain of custody, reporting).

Data Acquisition and Duplication

- Principles of forensic soundness (no modification of original data, repeatability, documentation).
- Types of acquisition: physical vs. logical, targeted vs. full-disk, live vs. dead (post-mortem) acquisition.
- Imaging methods: bit-stream disk imaging, partition imaging, memory (RAM) acquisition, mobile/cloud acquisition.
- Use and configuration of write blockers, hardware vs. software, and protection of source media.
- Common image formats and metadata (raw/dd, E01, AFF, hash values, logs).
- Handling special media: RAID, SSDs, encrypted volumes, virtual machines, cloud snapshots.
- Verification of acquired images (hashing, integrity checks) and maintaining chain of custody.
- Secure storage, documentation, and duplication of images for analysis and disclosure.